

# Knowledge Representation

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*Summer semester 2026*

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April 2026



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## Part I

# Foundation of Knowledge Representation



# CHAPTER 1

## Introduction to KR: Definition, Scope, Motivation (Use Cases), and History

### 1.1 | Introduction

#### What is Knowledge Representation?

**Definition 1.1 Knowledge Representation**

KR is a central problem in Artificial Intelligence that focuses on developing sufficiently precise notations for representing knowledge

The core problem with KR is that the real world with its complexity, ambiguity and dynamism can never be fully represented in a computer-based system. Nevertheless, intelligent systems must be able to function effectively using partial, incomplete and sometimes contradictory information. [4]

#### Why is Knowledge Representation needed?

**The problem of data meaninglessness**

To understand the need for KR, let's consider a practical scenario:

**Example 1.1** A student database.

A hypothetical database contains the following entries:

| Name        | Major   | CGPA | Research            | ... |
|-------------|---------|------|---------------------|-----|
| Alice Smith | CS      | 3.8  | Genomics            | ... |
| Bob John    | Biology | 3.8  | Oncology            | ... |
| Carol Li    | Zoology | 3.7  | Animal colonisation | ... |

Without knowledge representation, this data is disjointed. The entries exist in isolation across thousands of files. A database query system can only perform exact keyword matches:

- Query 1: "Show me all AI researchers" → No results (not explicitly mentioned in the data)
- Query 2: "Which students could work on this patent?" → No answer possible
- Query 3: "What are the emerging trends?" → No analysis possible

Data is meaningless without context and relationships. It does not allow for intelligent reasoning or deeper analysis.

What knowledge representation makes possible

With knowledge representation, we can structure the data and transfer isolated records into interconnected, queryable knowledge using:

- **ontologies**: Explicitly specify which concepts exist (student, researcher, field of research) and how they are categorised.
- Knowledge graphs: Link entities together: Alice → researcher → AI

This allows for semantic searches, automatic relationship discovery and reasoning across all connection.

**Example 1.2**

- Semantic search
  - “Find students researching machine learning?”
  - “Show me all AI researchers”: The system can use transitive relationships to find relevant students.
  - “Which students are a good fit for this patent?": The system can match skills and areas of interest.
- Automatic relationship discovery
  - “What are emerging research trends?”
  - Recognising patterns across all data
  - Identifying emerging trends through interconnected information

| Data                   | Information                                      | Knowledge                                                                                                                     |
|------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Raw, unprocessed facts | Structured and contextualised data               | Information combined with experience and reasoning rules                                                                      |
| e.g. CGPA: 3.8         | e.g. Alice has a CGPA of 3.8 in Computer Science | e.g. Alice could work on this AI patent because she has a strong foundation in computer science and is interested in genomics |

Table 1.1: Data vs. Information vs. Knowledge

KR enables machines to make informed decisions with fewer rules.

KR Perspective

**Traditional ML** systems follow a black-box approach which means the decision-making process is not transparent and the reasoning behind predictions is not accessible.

- Generalisation: Requires continuous retraining with new data
- Explainability: Difficult or impossible to explain decisions
- Data flexibility: Highly dependent on large amounts of training data

**Knowledge-Based AI** utilises structured, **explicit knowledge**.

**Definition 1.2** **explicit knowledge**  
Knowledge that is formally encoded in symbols, rules or graphs and can be processed by machines

- Generalization: explicit knowledge transfers to new problems without retraining from scratch[3]
- Explainability: encoded knowledge enables transparent reasoning and human-understandable decisions[1]
- Data flexibility: Modifiable

## Representation Schemes

### Definition 1.3 representation scheme

A representation scheme is a formal notation used to define a knowledge base (Mylopoulos 1980). A knowledge base contains facts and rules and serves as a model of the real world. A representation scheme is therefore the ‘language’ in which we express knowledge, much like programming languages define the syntax and semantics in which we write instructions.

**Example 1.3 Pac-Man ghost AI** has a knowledge-based approach: We code ghost behaviour as explicit rules.

1. IF player\_nearby THEN chase
2. IF frightened THEN flee
3. ELSE patrol

Decisions are transparent, explainable, and modifiable without recomputing. (Ogland 2016)

## Challenges in Knowledge Representation

Knowledge representation involves designing formal structures that allow computers to store, interpret, and reason about information and every design decision has advantages and caveats.

**Choosing the representation** is one of the design challenges. **Semantic networks** are intuitive and visually understandable but have limited inference. **First-order logic** on the other hand is expressive, but has computationally costly reasoning.

**Desired properties** can conflict. Systems should ideally be expressive enough to capture rich real-world knowledge while remaining efficient enough to reason in practical time. Tradeoffs are between expressiveness and tractability, and Human readability and machine efficiency.

**Application-driven constraints** arise from, different domains requiring different forms of representation. E.g. In natural language processing (NLP), systems must handle ambiguity and context. In medical diagnosis, representations must manage uncertainty, causal relationships, and strict correctness requirements. In question answering, systems need fast retrieval combined with reasoning over large amounts of structured and unstructured data

**The core idea of Knowledge Representation** is not just a data structure. It plays multiple roles simultaneously and those roles can conflict with each other:

- Surrogate for the world: Acts as a substitute for real world objects and relationships
- Ontological commitments: Defines what exists and how things are categorized
- Theory of reasoning: Provides foundation for logical inference and deduction
- Medium for efficient computation: Enables performant computational processing
- Medium of Human expression: Allows humans to express knowledge in understandable ways

## 1.2 | Taxonomy of Representation Schemes

Representational schemata can be divided into four basic categories:

- **Declarative representations**

- Logical representation schemes
- Semantic (network) schemes
- Procedural representations
- Hybrid approaches
- Frame-based representations

## Declarative representations

### Definition 1.4 declarative representation

Declarative representations describe facts about the world as logical statements

**Declarative representations** focus on truth and semantics and enable deductive reasoning. The schema is independent of how the knowledge is used.

There are two subtypes:

### 1. Logical representation schemes

#### Definition 1.5 logical representation scheme

Representation schema based on formal logic and represents knowledge through logical formulas. It is strongly based on semantics and inference

#### Example 1.4 First order Logic



**Figure 1.1:** This is Tweety. What can we infer about Tweety with the information given below? (image from <https://www.pexels.com/photo/pink-cockatoo-perched-in-natural-habitat-29732587/>)

We know that all bird can fly if they are not flightless birds, here generalized as penguins. We know that Tweety is a bird and not a penguin.

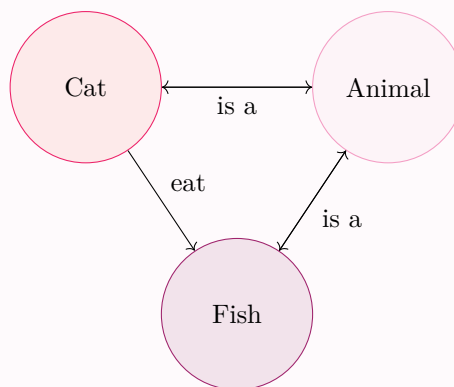
Fore formal:

Birds can fly:  $\forall x (Bird(x) \wedge \neg Penguin(x) \rightarrow CanFly(x))$   
 Facts:  $Bird(Tweety)$  and  $\neg Penguin(Tweety)$   
 Therefore we can infer:  
 $CanFly(Tweety)$

## 2. Semantic (network) schemes

### Definition 1.6 semantic representation scheme

Graph-based representations in which nodes represent entities and edges represent relationships between them. It supports inheritance and classification



**Figure 1.2:** Semantic Network

### Example 1.5

## Procedural representations

### Definition 1.7 procedural representation

Representation schema that encodes knowledge as procedures or programmes that describe how to use that knowledge. It is efficient but hard to interpret



**Figure 1.3:** This is Dr. Dokter. She wants to diagnose her patient. (image from <https://www.pexels.com/photo/female-doctor-in-scrub-suit-7659874/>)

#### Example 1.6

## 1.3 | Distinguishing Features of KR Systems

Defis: Frame, Inference,

kinds of reasoning inference types (23+) David, Shrobe and Szolovist

Wie man mit Incompleteness umgeht

## 1.4 | Applications of Knowledge Representation

Enterprise Knowledge Management

## 1.5 | Future Directions

Multi-hop reasoning Unified framework

knowledge Aggregation

def attention weight

## 1.6 | Course structure

foundation of KR

Knowledge graphs

KR in mL

grading:

50% of the exercise points (full portfolio) (Abgabe in 3er Gruppen(?)) present one exercise in tut present one paper in seminar 40–45 min (3er gruppen) final report 5–10 pages (50% marks on final report)

Es wird ein Moodle geben

## 1.7 | Formalities



# CHAPTER 2

## Logical Foundations: First-Order Logic (FOL)

Knowledge representation requires a formal system that expresses knowledge precisely and derives consequences from that knowledge.<sup>[2]</sup>

The most relevant formal calculations are **propositional Logic** and **first-order logic**.

### Definition 2.1    **propositional Logic**

todo

### Example 2.1    “If a person is rich, they have a nice car”

That statement becomes

$$\begin{aligned} BobIsRich &\Rightarrow BobHasNiceCar \\ JaneIsRich &\Rightarrow JaneHasNiceCar \\ &\vdots \end{aligned}$$

Having to define every rule for every person is impractical and unintuitive though.

## 2.1 | Syntax

6 - Objects sind x,y, ... 8 - not a well-formed formula weil es keinen Sinn macht

14 - unary predicate, welche gibt es noch??

### Definition 2.2    **formula**

todo

### Definition 2.3    **atomic formula**

todo

### Definition 2.4    **compound formula**

todo

### Definition 2.5    **quantified formula**

todo

### Example 2.2    **Building a Formula – Example from OLP**

### Definition 2.6    **free**

todo

**Definition 2.7**    **bound**

todo

**Example 2.3**    Von f 18

**Definition 2.8**    **sentence**

todo

**Example 2.4**    Von f 19

**Definition 2.9**    **substitution**

todo

**Example 2.5**    Von f 20.1

**Definition 2.10**    **captured**

todo

**Example 2.6**    Von f 21

## 2.2 | Semantics

**Definition 2.11**    **language structure**

todo

**Definition 2.12**    **domain**

todo

**Definition 2.13**    **valid**

todo

**Definition 2.14**    **satisfiable**

todo

**Definition 2.15**    **unsatisfiable**

todo

## 2.3 | Inference

**Definition 2.16**    **inference**

todo

**Definition 2.17**    **Modus Ponens**

todo

**Definition 2.18**    **Modus Tollens**

todo

**Definition 2.19**    **Universal Instantiation**

todo

**Definition 2.20**    **Existential Instantiation**

todo

**Definition 2.21**    **Prenex Normal Form**

todo

F 38 in step 1: Rule 1:  $A \rightarrow B = \neg A \cup B$ Rule 2:  $\neg \forall x(P(x)) \equiv \dots$  Ist es hier neg oder sim ??**Definition 2.22**    **clause**

todo

**Definition 2.23**    **literal**

todo

**Definition 2.24**    **Conjunctive Normal Form**

todo

**Definition 2.25**    **Resolution-based prover**

todo

**Definition 2.26**    **resolution**

todo

## 2.4 | Proof Systems & Completeness

**Definition 2.27**    **Natural Deduction**

todo

**Definition 2.28**    **Sequent Calculus**

todo

**Definition 2.29**    **derivability relation**

todo



# CHAPTER 3

## Logic Programming and Automated Reasoning



# CHAPTER 4

## Structured Knowledge Representation



## Part II

# Knowledge Graphs



# CHAPTER 5

Start into semantic networks and knowledge graphs reasoning



# CHAPTER 6

## The Resource Description Framework (RDF)



# CHAPTER 7

## Knowledge Graph Reasoning



# CHAPTER 8

## Knowledge Acquisition from Data



# CHAPTER 9

## Representation Learning



## Part III

# Knowledge Representation in Machine Learning



# CHAPTER 10

## Knowledge Graph Embeddings



# CHAPTER 11

## Graph Neural Networks for Knowledge Representation



# Part IV

## Applications



# CHAPTER 12

## Knowledge Tracing



# CHAPTER 13

## Knowledge in Large Language Models



# CHAPTER 14

## KG- Aware Applications



# Backmatter



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# Definitions

**atomic formula** todo. 15

**bound** todo. 16

**captured** todo. 16

**clause** todo. 17

**compound formula** todo. 15

**Conjunctive Normal Form** todo. 17

**data** todo. 8

**declarative representation** Declarative representations describe facts about the world as logical statements. 9, 10

**derivability relation** todo. 17

**domain** todo. 16

**Existential Instantiation** todo. 17

**explicit knowledge** Knowledge that is formally encoded in symbols, rules or graphs and can be processed by machines. 8

**first-order logic** todo. 9, 15

**formula** todo. 15

**frame-based representation** todo. 10

**free** todo. 15

**inference** todo. 16

**information** todo. 8

**knowledge** todo. 8

**knowledge base** todo. vi, 9

**Knowledge Representation** KR is a central problem in Artificial Intelligence that focuses on developing sufficiently precise notations for representing knowledge. 7

**language structure** todo. 16

**literal** todo. 17

**logical representation scheme** Representation schema based on formal logic and represents knowledge through logical formulas. It is strongly based on semantics and inference. 10

**Modus Ponens** todo. 16

**Modus Tollens** todo. 16

**Natural Deduction** todo. 17

**ontology** todo. 8

**Prenex Normal Form** todo. 17

**procedural representation** Representation schema that encodes knowledge as procedures or programmes that describe how to use that knowledge. It is efficient but hard to interpret. 10, 11

**propositional Logic** todo. 15

**quantified formula** todo. 15

**representation scheme** A representation scheme is a formal notation used to define a **knowledge base** (Mylopoulos 1980). A knowledge base contains facts and rules and serves as a model of the real world. A representation scheme is therefore the ‘language’ in which we express knowledge, much like programming languages define the syntax and semantics in which we write instructions.. 9

**resolution** todo. 17

**Resolution-based prover** todo. 17

**satisfiable** todo. 16

**semantic network** todo. 9

**semantic representation scheme** Graph-based representations in which nodes represent entities and edges represent relationships between them. It supports inheritance and classification. 11

**sentence** todo. 16

**Sequent Calculus** todo. 17

**substitution** todo. 16

**Universal Instantiation** todo. 17

**unsatisfiable** todo. 16

**valid** todo. 16

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